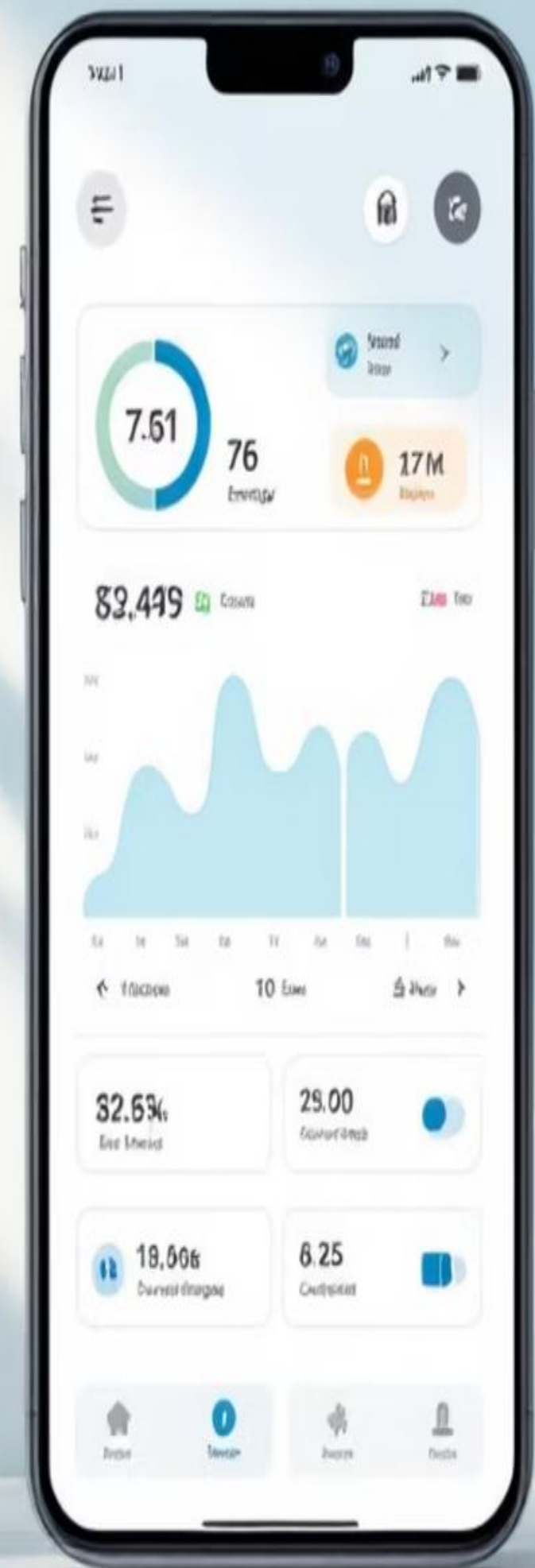


IoT-Based Smart Energy Consumption Monitor

An affordable, real-time solution for tracking electricity usage and reducing costs

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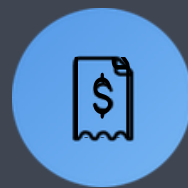
The Energy Monitoring Challenge

Most homeowners and small businesses lack visibility into their electricity consumption until it's too late.



No Real-Time Data

Traditional meters only show cumulative usage with no immediate feedback on consumption patterns



Monthly Bill Surprises

Users receive bills at month-end with no opportunity to adjust usage mid-month



Hidden Waste Sources

Difficult to identify which appliances or devices are consuming excessive power

Why Real-Time Monitoring Matters

Spot energy hogs instantly

Identify appliances that are driving up your electricity bill

Adjust behavior while it matters

Make immediate changes to reduce consumption and save money

Prevent costly surprises

Track usage throughout the month instead of waiting for the bill



Project Objectives

01

Real-Time Monitoring

Track electricity consumption continuously with instant feedback on voltage, current, and power levels

02

Precise Measurement

Calculate accurate voltage, current, and power values for detailed energy analysis

03

Cost Estimation

Estimate energy usage costs (₹/kWh) for individual appliances and total consumption

04

Low-Cost IoT Solution

Develop an affordable monitoring system accessible to homeowners and small businesses



Literature Review: Current State

Traditional Energy Meters

Provide only total consumption readings without real-time insights or appliance-level breakdown

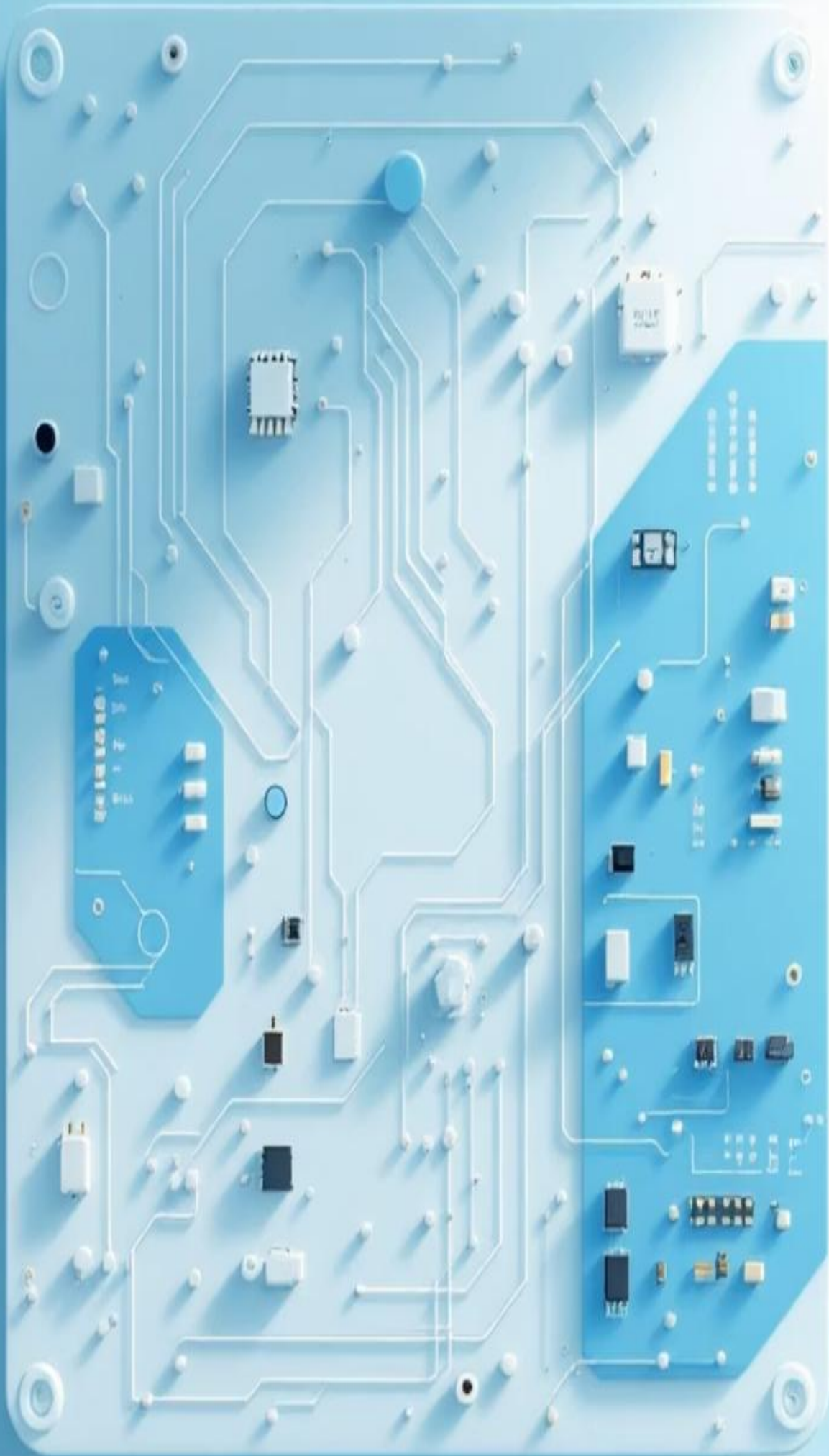
IoT-Based Systems

Use NodeMCU and sensors for real-time monitoring with cloud access and data visualization

Existing Capabilities

Offer tracking features, alerts, and dashboards but focus primarily on energy (kWh) without cost analysis

PAPER	METHOD USED	FEATURES	LIMITATIONS
Kumar et al. (2023)	IoT (ESP8266)	Low-cost + real-time monitoring	• No cost calculation • Limited accuracy
Pradhan et al. (2023)	ESP8266 + Sensors	Measures V, I, Power	• No cost estimation • Internet dependency
Hossain et al. (2023)	IoT + Cloud	Data analytics + trends	• Complex system • Non user-friendly
Verma et al. (2023)	IoT + Relay	Monitoring + load control	• No cost insights • No appliance-level monitoring
Karthik et al. (2023)	Wireless Sensors	Real-time transmission	• Limited scalability • Maintenance difficulty
Singh et al. (2024)	IoT Smart Meter	Real-time monitoring + mobile app	• No appliance-level cost • Internet dependency
Lee et al. (2024)	Smart Energy System	AI-based analysis	• High complexity • Non user-friendly
Zhang et al. (2025)	IoT + AI	Smart Automation	• Expensive • Difficult to maintain



Research Gap & Innovation



Research Gap

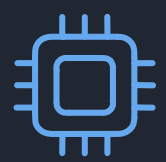
No simple, low-cost method exists to calculate appliance-level electricity costs in real-time



Our Innovation

Real-time cost per unit calculation (₹/kWh) using an affordable, accessible setup

System Components



NodeMCU (ESP8266)

Main controller with built-in Wi-Fi for IoT connectivity and data transmission



ACS712 Current Sensor

Measures electrical current flowing through circuits with high accuracy



Voltage Sensor Module

Monitors voltage levels to calculate power consumption accurately



16×2 LCD Display

Shows real-time readings via I2C interface for local monitoring



Relay Module

Controls electrical loads remotely based on programmed conditions



5V Power Supply

Provides stable power to all system components and sensors



How the System Works

1

Measure

Sensors continuously monitor voltage and current from connected appliances

2

Calculate

NodeMCU processes measurements to determine power and energy consumption

3

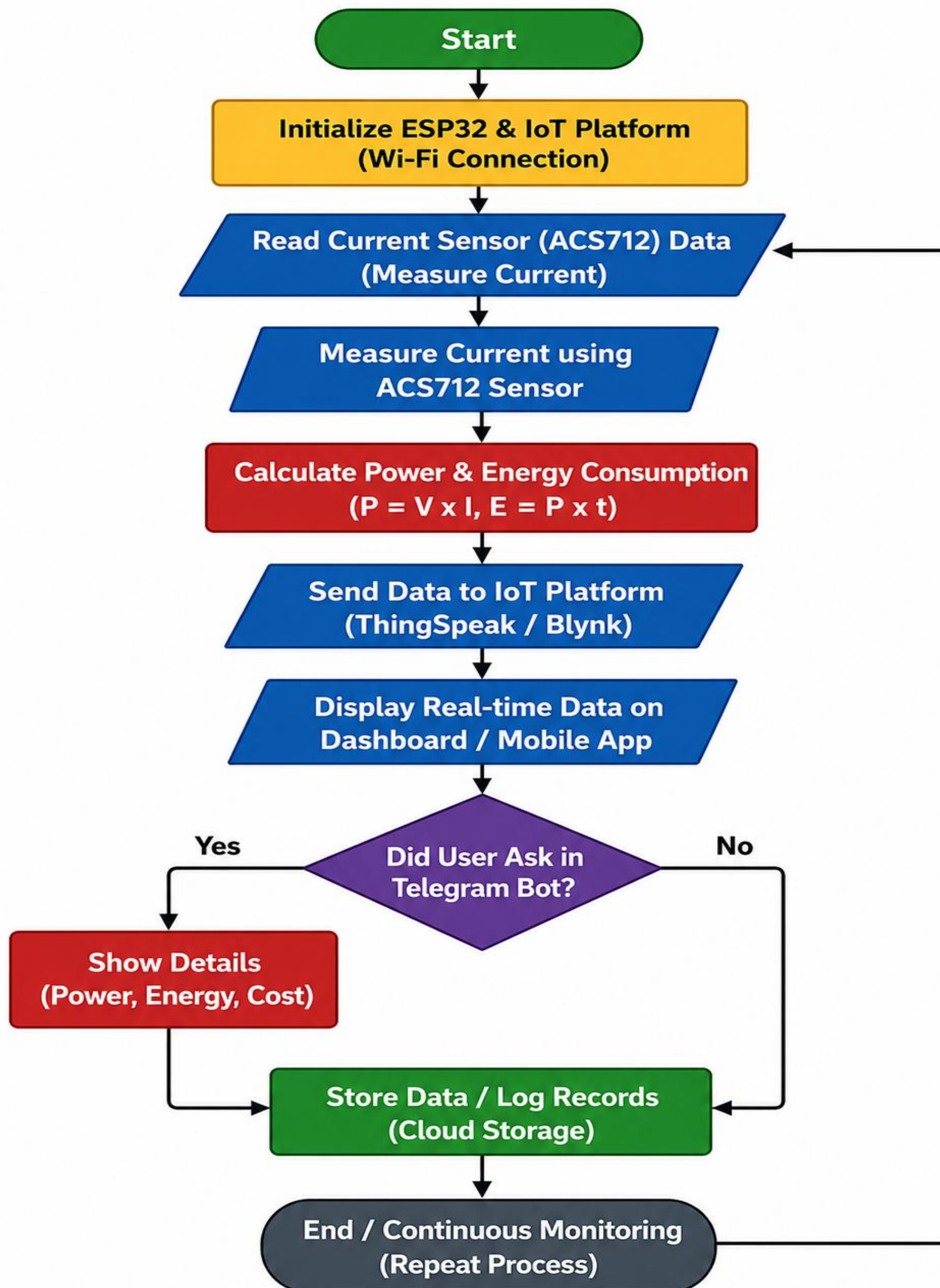
Display

Real-time readings shown on LCD display and transmitted via Wi-Fi

4

Analyze

Users access data remotely to identify usage patterns and cost trends



Expected Benefits

Real-Time Insights

Continuous monitoring of voltage, current, power, and energy consumption with instant visibility

Remote Tracking

Access electricity usage data from anywhere using IoT-enabled devices and cloud platforms

Cost Calculation

Low-cost per-unit cost analysis (₹/kWh) for individual appliances and total household consumption

Waste Reduction

Identify and eliminate energy wastage to reduce electricity bills significantly

Affordable Solution

Efficient, user-friendly system built with low-cost components accessible to everyone

Implementation Results

1

Real-Time Monitoring Achieved

System successfully tracks voltage, current, power, and energy consumption with minimal latency

2

Remote Access Enabled

IoT connectivity allows users to monitor electricity usage from smartphones or computers

3

Cost Analysis Implemented

Real-time ₹/kWh calculation provides appliance-level cost insights for informed decisions

4

Waste Reduction Confirmed

Users can identify energy hogs and adjust behavior to reduce electricity bills effectively

5

Low-Cost Solution Delivered

Affordable, efficient, and user-friendly system built with accessible components